EAST WEST UNIVERSITY

Physics Lab

Department of MPS

Course Name: Engineering Physics -1

Course Code : PHY 109 LAB

Section No : 08

Group No : 05

Experiment No : 04

Name of the Experiment: To determine the wavelenght of various spectral lines by a spectrometer using a plane diffraction grating.

Date of allocation : 12.11.24

Date of submission : 19.11.24

Submitted To : Md. Hedayet Ullah

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19.11.24

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Experiment NO. 04

Name of the Experiment: To determine the wavelengths of various spectral lines by a spectrometer using a plane diffraction ~~gratting~~ grating.

Theory:

- When light passes through a narrow slit then light spreads out, this phenomenon is called diffraction.
- When wavelength, λ is compareable with the slit's width, d, then diffraction becomes pronounced, ~~d=~~ $d \approx \lambda$.
- The diffraction grating, a useful device for analyzing light sources, consists of a Large number of equally spaced parallel slits.

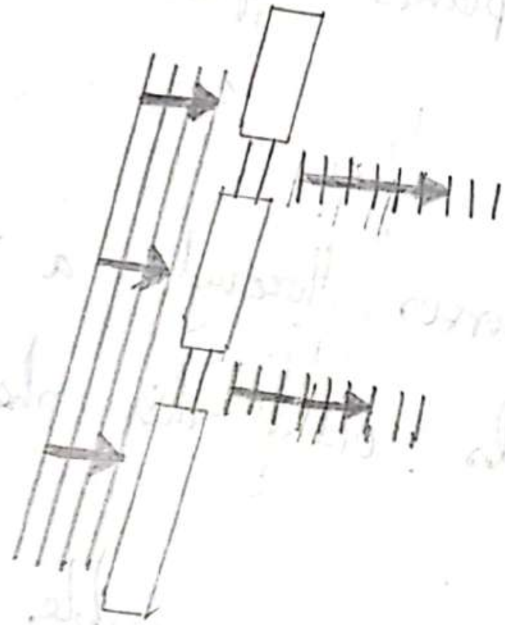


Figure : Diffraction (spreading) is not pronounced

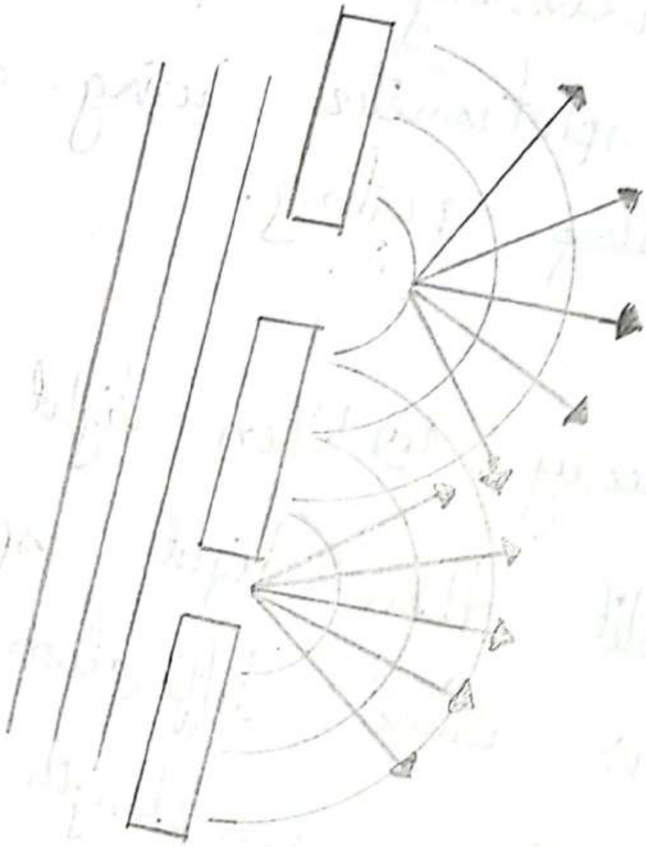


Figure : Diffraction not pronounced.

If a grating is ruled with 5000 ~~lines~~ lines/cm then it has a slit spacing

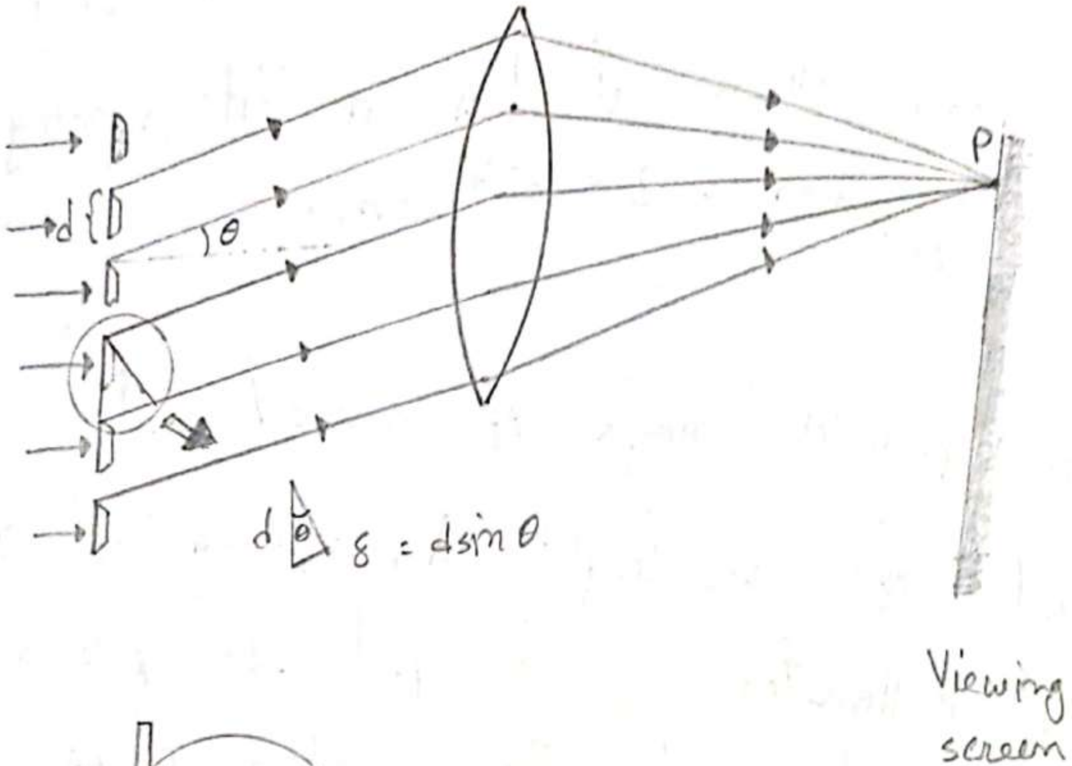
$$d = \frac{1}{5000} cm = 2 \times 10^{-4} cm.$$

If parallel rays of monochromatic light of wevelenght λ, coming out of the collimator of a spectrometer falls normally on a plane diffraction grating placed vertically on a prism table then a series of diffracted image of the collimator slit will be seen on both sides of direct image.

Parallel rays of a monochromatic light of wavelenght λ are incident on a diffraction grating in which the slit seperation is $(a+b)$. If the grating has N lines per unit lenght, the the grating spacing is given by :

$$a+b = \frac{1}{N} \quad \ldots\ldots (1)$$

C



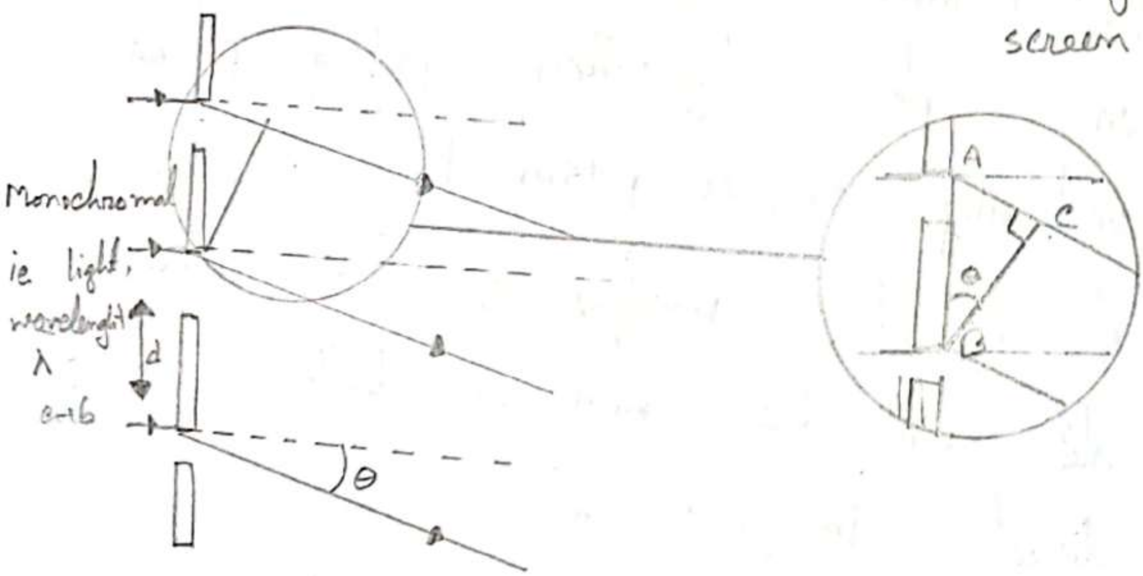


figure : Side view of a diffraction gratting. The slit separation is d, and the path difference between adjacent slits is $d\sin\theta$

Constructive interference only occurs along a few precise directions, one of which is show in the diagram. Light from ~~a~~ A must be in phase ~~of~~ with light ~~o~~from B, and this can only happen when the path diffrence is a whole number of complete wavelengths (even number of half-wavelengths).

Path diffrence, $AC = n\lambda$ where $n = 0, 1, 2, 3 \ldots$

Therefore, $(a+b)\sin\theta = n\lambda$ .... (2)

where, θ is the angle of diffraction.

The term n is called the spectrum order. If $n = 1$, we have the first order diffraction maximum. $\sin\theta$ can never be greater than 1, so there is a limit to the number of spectra that can be obtained.

Thus $\frac{\sin\theta}{N} = n\lambda$ ...... (3)

$\therefore \lambda = \frac{\sin\theta}{nN}$ .... (4)

The wavelenght ~~of λ~~ λ of any unknown light can be found out with the help of equation (4).

Apparalus :

- Spectrometer
- Spirit Level
- Plane diffraction ~~level~~ grating.
- Discharges tube , etc.

Data Sheet: Diffraction grating experiment:

$1\ inch = 2.54\ cm.$ $\quad\quad$ $1\ m = 100cm = 10^9 nm$

$$Vernier\ ConstAnt\ (V.C.) = \frac{Value\ of\ the\ smallest\ division\ in\ Main\ scale}{Total\ number\ of\ divisions\ in\ Vernier\ scale} = \frac{1}{60}$$

Grating constant, N = the number of lines or rulings per nanometer of the grating surface

$= \frac{15000}{2.54}$

Number of order, n	Description of the line	Reading for the angle of Diffraction θ								
		Left Side			Right Side					
		Main scale reading (S) in degree	Vernier scale reading (V) $V = V.C. \times Div$, in degree	Total reading (L) $L = S + V$ in degree	Main scale reading (S) in degree	Vernier scale reading (V) $V = V.C. \times Div$, in degree	Total reading (R) $R = S + V$ in degree	$2\theta = L \sim R$ in degree	θ in degree	Wave length $\lambda = \frac{Sin\theta}{nN}$ nm / m
1st order	Violet	70	0.217	70.217	100	0.067	100.067	29.85	14.925	4.36×10^{-5}
	Indigo	71.5	0.067	71.567	100.5	0.123	101.623	30.116	15.058	4.40×10^{-5}
	Blue	69	0.20	69.2	101	0.083	101.083	31.883	15.94	4.65×10^{-5}
	Green	68	0.25	68.25	102	0.05	102.05	33.8	16.9	4.92×10^{-5}
	Yellow	66	0.2	66.2	106	0.1	106.1	39.9	18.45	5.78×10^{-5}
	Orange									
	Red	65	0.15	65.15	110	0.05	110.05	44.9	22.35	6.60×10^{-5}
2nd order	Violet									
	Indigo									
	Blue									
	Green									
	Yellow									
	Orange									
	Red									

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Calculation: Wavelength, $\lambda = \dfrac{Sin\theta}{nN}$ = ...

	Violet	Indigo	Blue	Green	Yellow	Orange	Red
Standard value of Wavelength of different Color of **Sodium Light** (mainly **yellow color**)Source *nm*	447	450	471	502	588	620	668
Standard value of Wavelength of different Color of **Mercury Light** (mainly **Violet color**)Source *nm*	405	436	-	546	578	615	-
Experimental Value of wavelength in *nm*	436	440	465	492	578	....	660

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Percentage difference (Violet color) = {(447 − 443) ÷ 447} × 100 = 2.46%

Percentage difference (Indigo color) = {(450 − 440) ÷ 450} × 100 = 2.22%

Percentage difference (Blue color) = {(471 − 465) ÷ 471} × 100 = 1.274%

Percentage difference (Green color) = {(502 − 492) ÷ 502} × 100 = 1.992%

Percentage difference (Yellow color) = {(588 − 578) ÷ 588} × 100 = 1.7%

Percentage difference (Orange color) = ...

Percentage difference (Red color) = {(668 − 660) ÷ 668} × 100 = 1.2

Results:

	Violet	Indigo	Blue	Green	Yellow	Orange	Red
Wavelength in *nm*	436	440	465	492	578	....	660

Results :

	Violet	Indigo	Blue	Green	Yellow	Orange	Red
Wavelenght in nm	436	440	465	492	578		660

Discussion :

- We must position the light source directly in front of the collimator slit. This will maximize the amount of light entering the system, resulting in a bright image
- We should align vertical cross-wire, or preferably the center of the cross wires, with the edge of the slit image. It is essential to eliminate parallax between the cross-wires and the image.
- To avoid backlash error, we must rotate the telescope's tangent screw consistantly in the same direction during final adjustments.
- We should minimize the slit width for optimal image resolution. Also, when taking readings, we must carefully note any zero crossing on the main circular scale to ensure accurate measurements.